UIUC Policy Agent

Is it really impossible to be both fast and accurate? Maybe not.

Presenter: Xujun Zhou

Why Do We Need This?

Hard to Find

Degree requirements are scattered across PDFs, old web pages, and conflicting resources.

Slow Advisors

Email or scheduling a meeting means waiting days—students need answers now.

AI Hallucinations

Generic AI answers lack citations and may provide inaccurate or outdated figures.

Key Technologies & Methods

ReAct Agent Architecture

Unlike if-then logic, the ReAct pattern reasons dynamically — it evaluates the result of each step before deciding what to do next.

This mirrors how a researcher works: look up → verify → respond.

Agent 1 — FAISS Vector Search

Semantic search over a local database of official UIUC pages (LAS, Engineering, iSchool).

Agent 2 — Live Web Search

Activates when local results are insufficient; uses Tavily to retrieve and parse real-time web pages.

Agent 3 — Claim Verification

Before stating any specific figure, performs a secondary LLM check against the source to prevent hallucination.

Workflow

```
(uiuc-rag) C:\Users\admin\Desktop\IS597AI\597FinalProject-xujunz2>python uiuc_policy_agent.py
UIUC Policy Agent
Type 'reset' to clear history, 'exit' to quit.

Select LLM backend:
  1. OpenAI  (gpt-5.4-mini)   - requires OPENAI_API_KEY
  2. Claude  (claude-opus-4-6) - requires ANTHROPIC_API_KEY
Enter 1 or 2 [default 1]: 2
[agent] Using backend: Claude (claude-opus-4-6)

You: What are the specific credit requirements for graduating from the BSIS program?
  [tool] query_local_index(query='BSIS credit requirements graduation', top_k=3)
[index] loading FAISS index and embedding model...
```

Workflow

```
[index] loaded 3 documents
  [tool] verify_claim(claim='The BSIS program requires a total of 123 hours of course credits, including 21 h
ours of required courses and 30 hours of iSchool electives.', source_text='The BSIS program includes seven re
quired courses as well as a variety of electives among different pathways to customize your degree. PLAN your
 BSIS degree A total of 123 hours of course credits are required: 21 hours of required courses 30 hours of iS
chool electives Includes a variety of topics across pathways general education requirements electives within
or outside the school, including minor(s) and/or certificate(s)')

Agent: Here are the specific credit requirements for graduating from the **BSIS (Bachelor of Science in Infor
mation Sciences)** program at the iSchool:

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### Total Credits Required: **123 hours**

The 123 hours break down as follows:

| Category | Credit Hours |
|---|---|
| **Required Courses** | 21 hours |
| **iSchool Electives** | 30 hours |
| **General Education Requirements** | Varies |
| **Free Electives** (within or outside the school, including minors/certificates) | Remaining hours |
```

Workflow

```
### Required Courses (21 hours – 7 courses):
1. **IS 100** – Introduction to the iSchool Core
2. **IS 101** – Introduction to Information Sciences
3. **IS 202** – Social Aspects of Information Technology
4. **IS 203** – Analytical Foundations for Information Problems
5. **IS 204** – Research Design for Information Sciences
6. **IS 205** – Programming for Information Problems *(or another programming course)*
7. **IS 206** – Introduction to Database Concepts & Applications

### Plus one "Information Sciences and Culture" course (select one):
- IS 308 – Race, Gender and Information Technology
- IS 309 – Computers and Culture
- IS 310 – Computing in the Humanities
- IS 311 – Historical Foundations of the Information Society

### iSchool Electives (30 hours):
These include a variety of topics across different **pathways**, allowing you to customize your degree.

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The remaining hours are filled with **general education requirements** and **electives** within or outside th
e iSchool, which can include minors and/or certificates.

📌 **Source:** [iSchool – BSIS Coursework](https://ischool.illinois.edu/academics/undergraduate/bs-informatio
n-sciences/coursework)

For personalized advising, you can contact the BSIS advising team at **bsis-advising@illinois.edu**.
```

Workflow

```
You: As an MSIM student, do credits earned from CS courses count toward graduation requirements? Is there a l
imit on the number of credits?
  [tool] query_local_index(query='MSIM credit requirements CS courses graduation', top_k=3)
  [tool] query_local_index(query='MSIM Master of Science Information Management degree requirements credit li
mits', top_k=3)
  [tool] search_web(query='UIUC iSchool MSIM degree requirements CS courses credit limit graduation')
  [tool] search_web(query='UIUC MSIM Master of Science Information Management coursework requirements outside
 courses')
  [tool] fetch_and_extract(url='https://ischool.illinois.edu/academics/graduate/ms-information-management/pla
n-your-program')
  [tool] fetch_and_extract(url='https://catalog.illinois.edu/graduate/is/information-management-ms/')
  [tool] verify_claim(claim='MSIM students may transfer up to 12 hours of graduate credit from other departme
nts at Illinois toward the 40 required credit hours.', source_text="Non-iSchool courses You may transfer up t
o 12 hours of graduate credit hours from several other departments at Illinois that can be applied to the 40
required graduate credit hours for the master's degree. Enrollment in other courses is subject to department
approval.")
  [tool] verify_claim(claim='The MSIM requires a minimum of 28 credit hours within the iSchool unit.', source
_text='Other Requirements may overlap Grad Other Degree Requirements Requirement Description Minimum Hours Re
quired within the Unit 28 Minimum 500-level Hours Required Overall 12 Maximum Guided Research/Independent Stu
dy 4 Competency in at least one programming language Minimum GPA: 2.75')
  [tool] verify_claim(claim='MSIM students must complete 40 graduate credit hours total to earn the degree.',
 source_text='Students complete 40 graduate credit hours to earn an MS in information management. All MSIM st
udents must complete the three required courses: IS 507 Data, Statistical Models, and Information (4 credit h
ours); IS 504 Sociotechnical Information Systems (4 credit hours); and IS 515 Information Modeling (4 credit
hours) or IS 455 Database Design and Prototyping (4 credit hours).')
```

Workflow

```
Agent: Great — all claims verified. Here's a comprehensive answer:

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## Can CS Courses Count Toward the MSIM Degree? — Yes, With Limits

### Total Degree Requirement
The **MSIM (MS in Information Management)** requires **40 graduate credit hours** to graduate.

### Non-iSchool Courses (Including CS)
Yes, **credits earned from CS courses can count toward graduation**, but there is a cap:

> **You may apply up to 12 hours** of graduate credit from other departments at Illinois (including CS) towar
d the 40 required credit hours.

Enrollment in non-iSchool courses is **subject to department approval**.

### Minimum Hours Within the iSchool
A minimum of **28 credit hours must be completed within the iSchool unit**. This is why the outside-course li
mit is effectively 12 hours (40 total - 28 within unit = 12 outside).
```

Workflow

```
### Quick Breakdown

| Requirement | Hours |
|---|---|
| **Total credit hours** | 40 |
| **Required iSchool courses** (IS 507 + IS 504 + IS 515 or IS 455) | 12 |
| **Elective courses** | 28 |
| **Minimum hours within the iSchool** | 28 |
| **Maximum hours from outside departments (e.g., CS)** | **12** |
| **Minimum 500-level hours overall** | 12 |
| **Maximum Guided Research/Independent Study** | 4 |
| **Minimum GPA** | 2.75 |

### Key Takeaways
1. ✅ CS graduate courses **do count** toward your 40-hour requirement.
2. 🔒 The limit is **12 credit hours** from non-iSchool departments.
3. 📋 You need **department approval** to enroll in outside courses.
4. 🏫 At least **28 hours** must come from iSchool (IS-prefix) courses.

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📌 **Sources:**
- [iSchool - MSIM Plan Your Program](https://ischool.illinois.edu/academics/graduate/ms-information-managemen
t/plan-your-program)
- [UIUC Course Catalog - Information Management, MS](https://catalog.illinois.edu/graduate/is/information-man
agement-ms/)
```

Areas for Improvement & Future Plans

01

Automatic Database Populating

Replace manual URL entry with a workflow that automatically discovers, crawls, and re-indexes official UIUC resources — keeping the knowledge base always up to date.

02

Front-End User Interface

Move beyond the command line. Build a clean, browser-based chat interface that any student can open without technical setup.

03

Broader Coverage

Expand the local index from 3 colleges to all UIUC colleges — and eventually to multiple universities — with accuracy validation at each stage.

Thank You

If you have any questions, feel free to ask!